SmartLocateSG

Singapore Site Selection intelligence

Group 13 (SCEZ - Group 2)

Product Intro • Architecture • Demo

CHOO **JUNG KIT**

ETHAN VINORAJ

MARCUS LEE JIA RONG

NEO **YU HUI** (LIANG YUHUI)

WAI YAN MIN KO KO

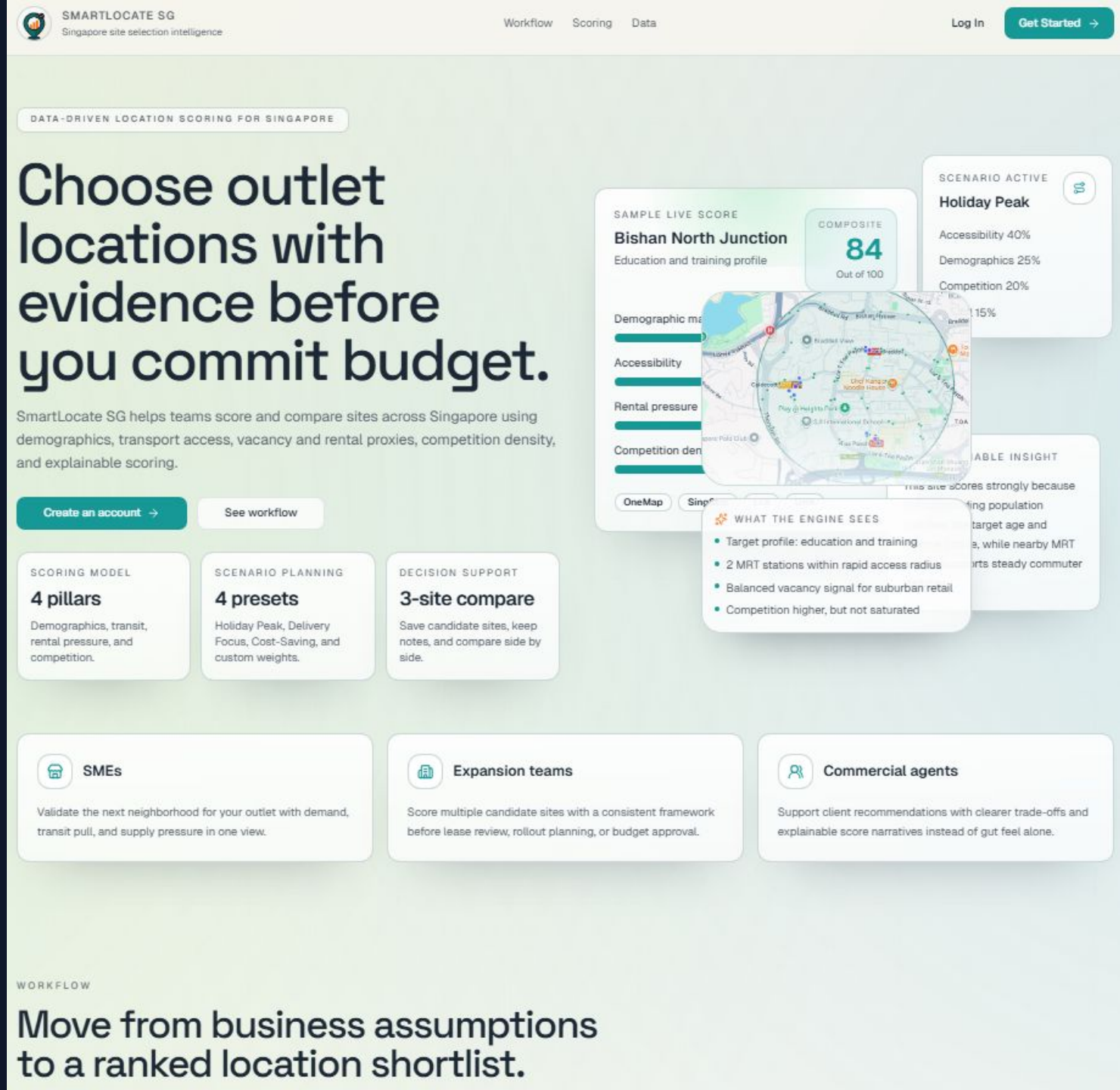
VALUE PROPOSITION

The Problem

Current business site selection **lacks unified scoring**. Businesses face **manual**, **fragmented research** processes that lead to **inefficient decision-making**.

The Solution

Evidence-based **site comparisons reduce research time** significantly. Users get **ranked**, **explainable scores** to inform critical decisions.



APPLICATION OVERVIEW

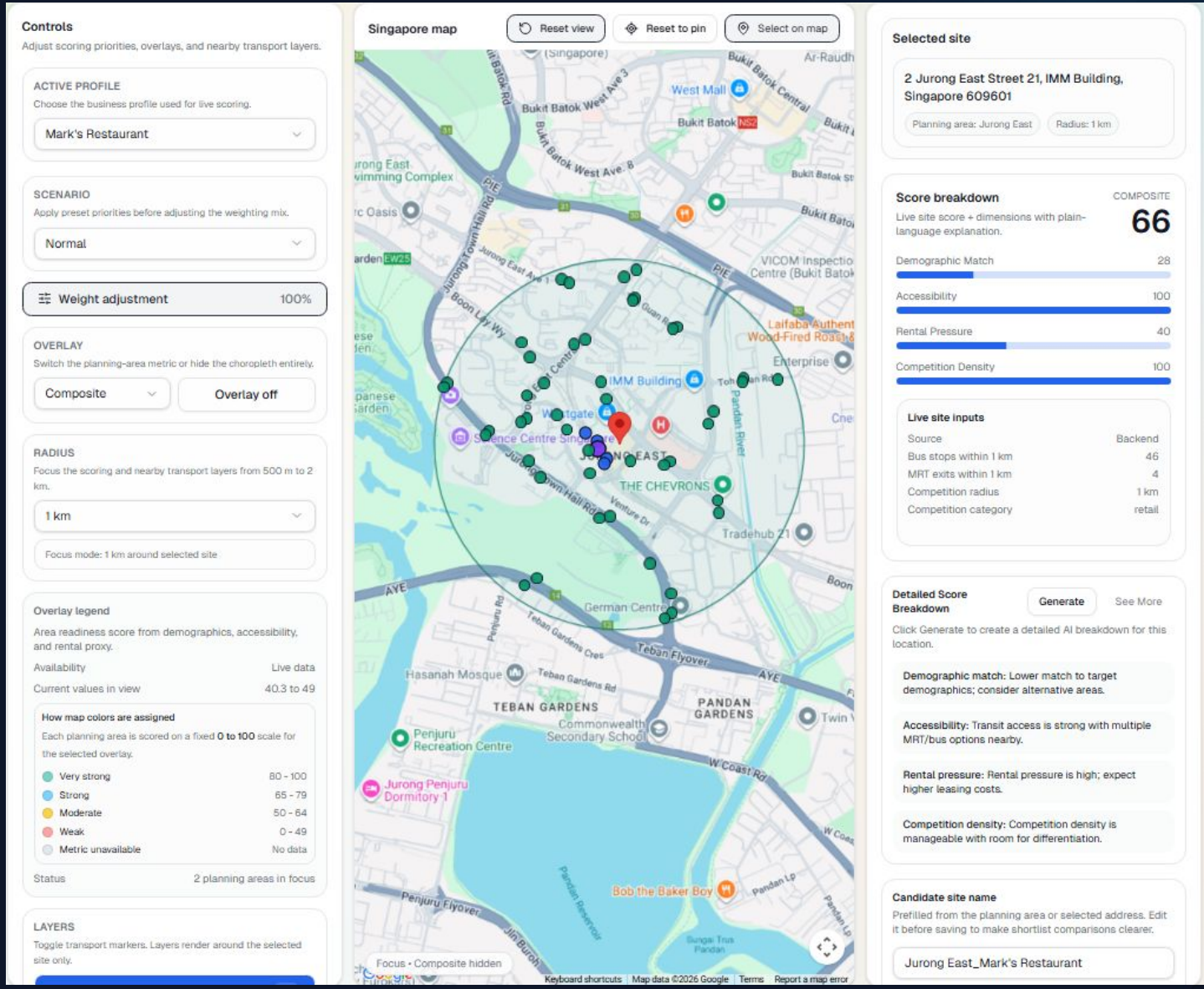
Location Scoring Prototype

A consolidated dashboard for data-driven site selection in Singapore.

Business Profiles: Manage scenario presets and weights.

Interactive Maps: Drop candidate pins and view breakdowns.

Site Comparison: Side-by-side analysis of top locations.



TARGET USER PERSONAS

Expansion Teams

Need deep data integration and comparative tools to justify high-value commercial expansions..

Small Business Owners (SMEs)

Need quick viability checks and simple, explainable scoring to launch their first physical outlet.

CORE FEATURE FLOWS

Profile Setup

Select or create a Business Profile to define weightage for scoring criteria.

Map Discovery

Drop pins on the map to calculate composite scores using the ScoreEngine.

Analysis

Save candidate sites, add qualitative notes, and compare up to 3 locations.

Transparency

View detailed score breakdowns for each location factor (Demographics, Accessibility, Competition Density).

SE Practices & Upgrade Readiness

Layered frontend structure

(pages/components/hooks/lib)
Improves reuse, testing surface, and onboarding for new contributors.

Resilience

API cache + mock/demo data + graceful AI fallback when Gemini unavailable, preventing hard failures.

Maintainability

Consistency via shared schemas/types (`shared/schema.ts`) and typed clients, reducing data-contract drift across app and APIs.

Operability

Admin tools for data/source status and demo reset, plus centralized utilities for storage and routing.

TECH STACK & DEPLOYMENT

React + Vite + TypeScript

Frontend

React + TypeScript + Vite

Backend

Node / Express API

Language

Strict TypeScript Typing

Database

Supabase

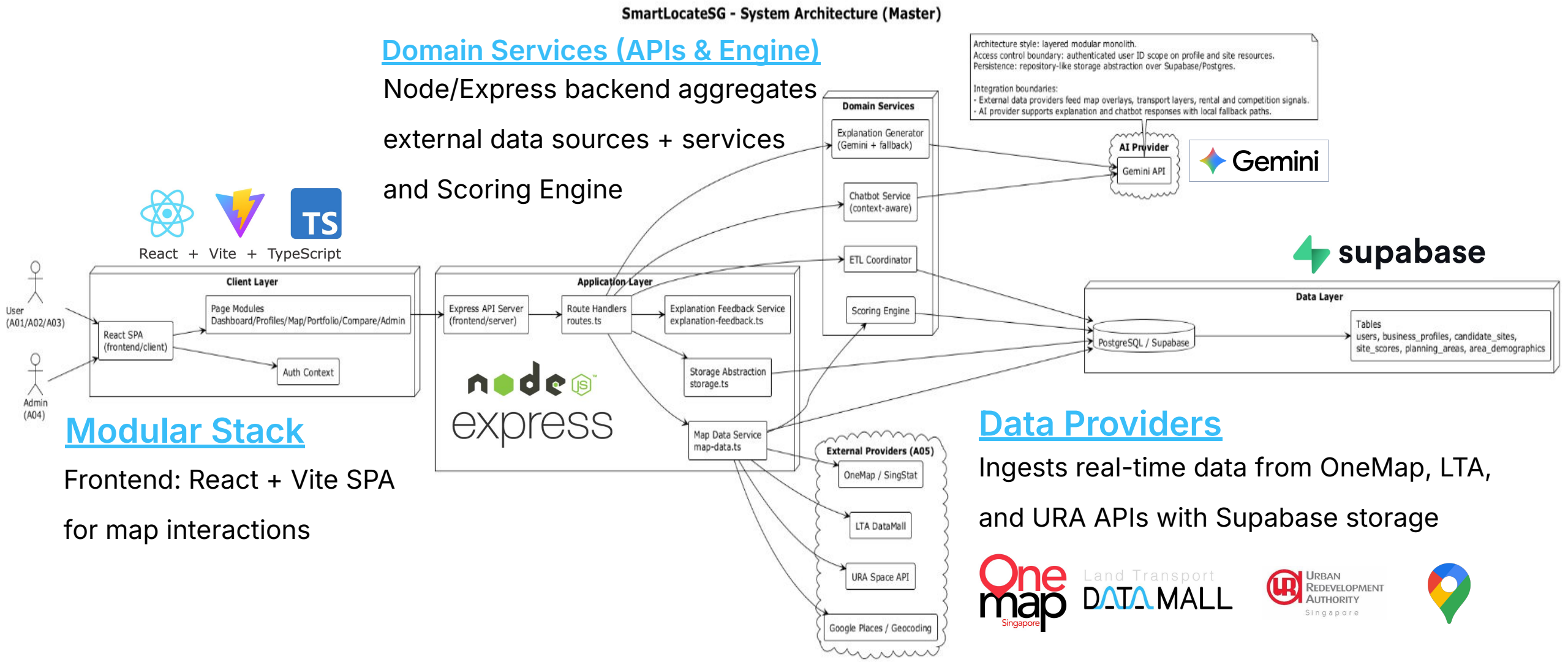
Chatbot (API)

Gemini-3.1-flash-lite

External Data (API)

OneMap + LTA + URA + Google Maps

SYSTEM ARCHITECTURE



Domain Services (APIs & Engine)

Node/Express backend aggregates external data sources + services and Scoring Engine

Modular Stack

Frontend: React + Vite SPA for map interactions

Data Providers

Ingests real-time data from OneMap, LTA, and URA APIs with Supabase storage

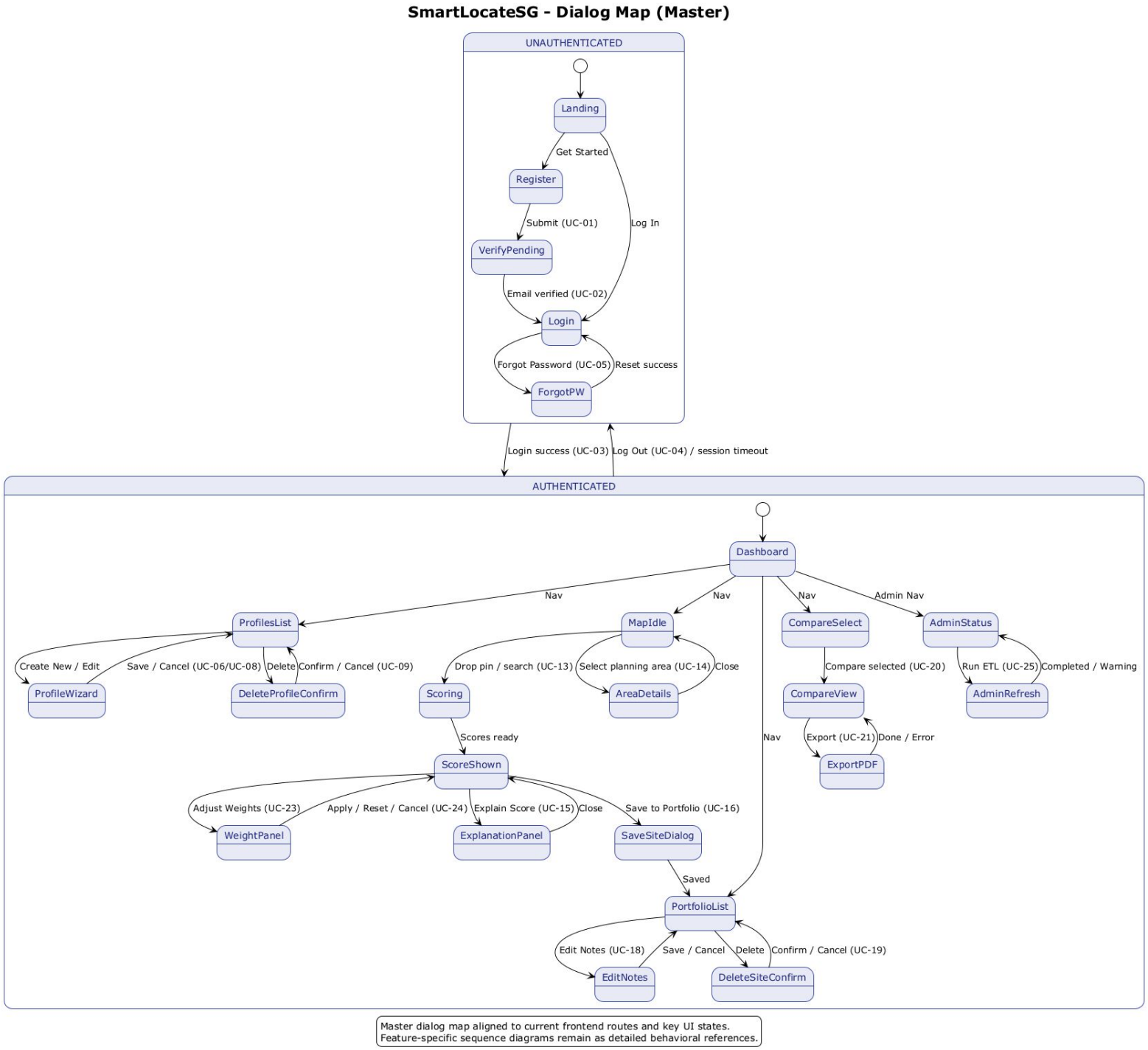
OneMap Singapore · Land Transport DATAMALL · Urban Redevelopment Authority Singapore

DIALOG MAP

Intelligent Guidance

Showcases the Flow of a typical User Interaction

- ✓ **Main Flows:** Guidance from profile creation to map exploration.
- **Fallbacks:** Intent clarification and contextual help links.



📂 diagrams\Dialog Map

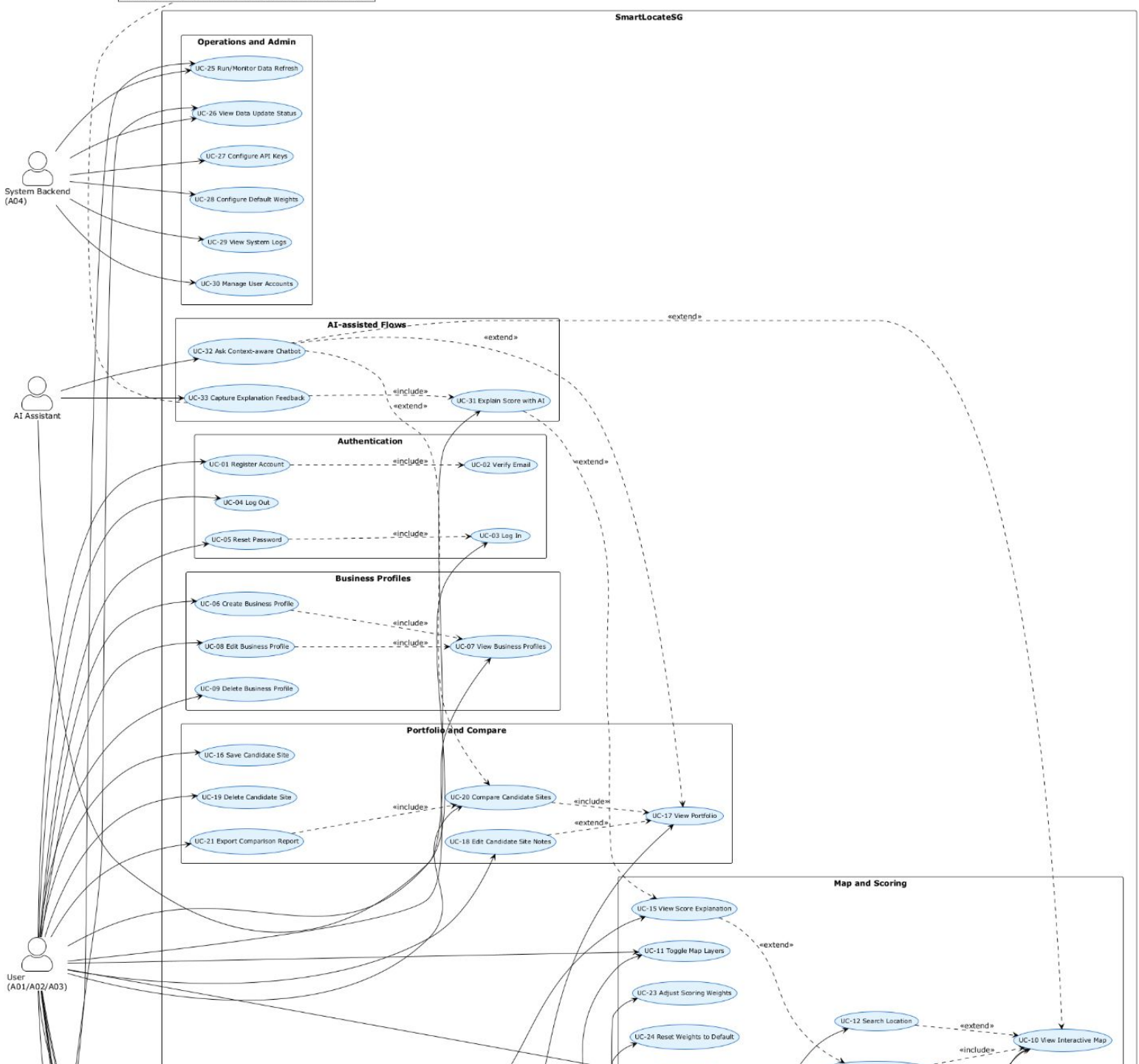
USE CASE DIAGRAM

Total: 33 Use Case Diagrams

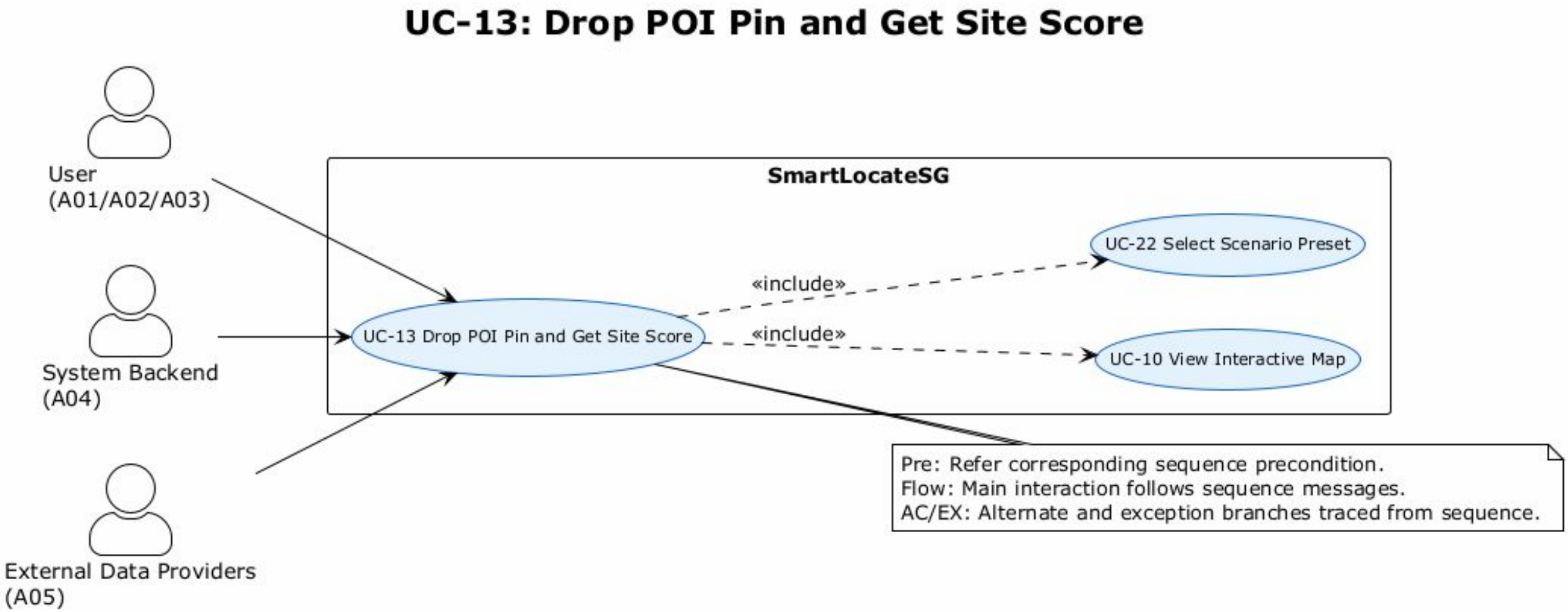
Showcases the relationships between actors and the primary functions (use cases).

- Operations & Admin
- Authentication
- Business Profiles
- Portfolio and Compare
- Map and Scoring
- AI-assisted Flow

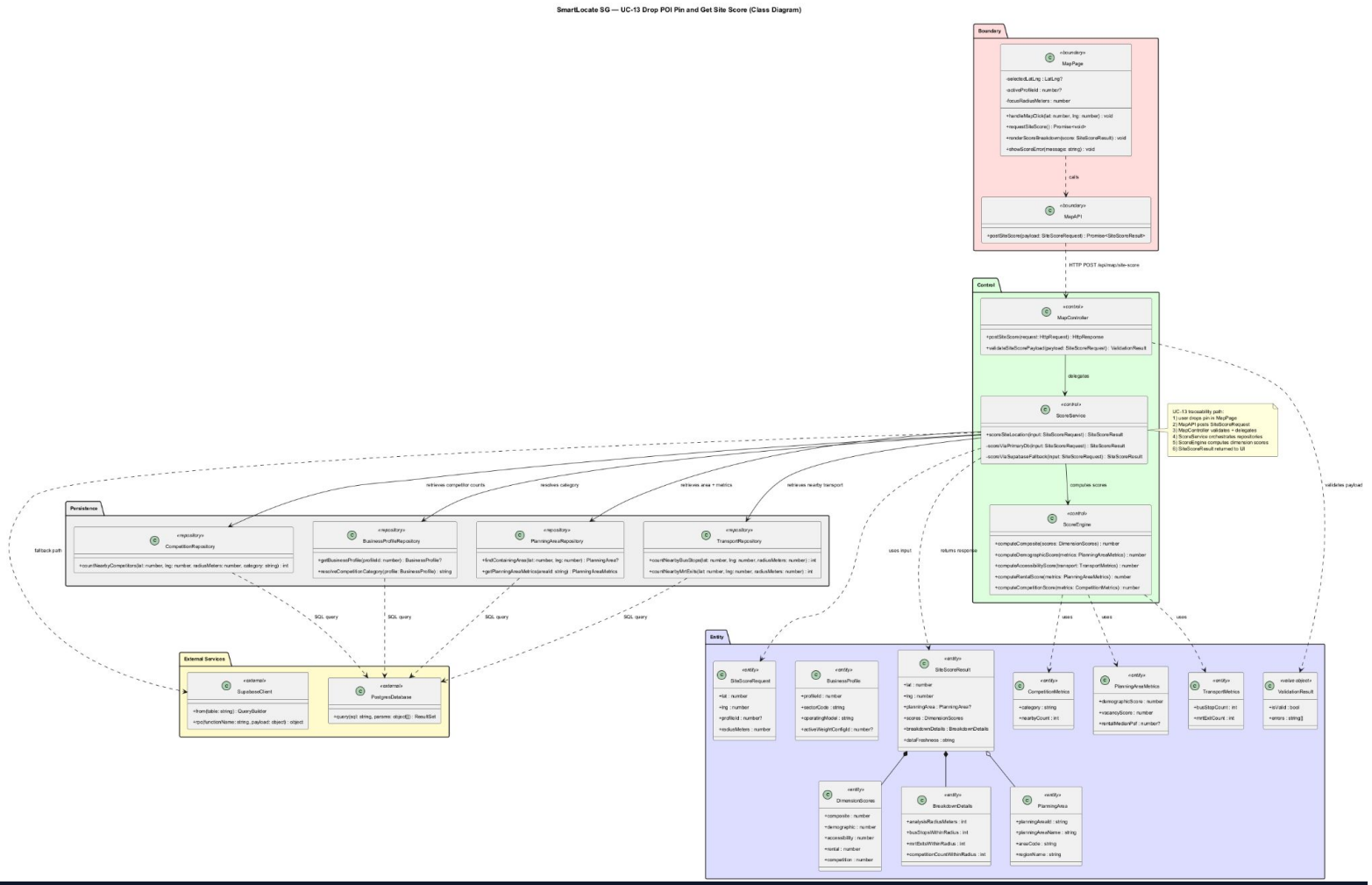
diagrams\Use Case Diagrams

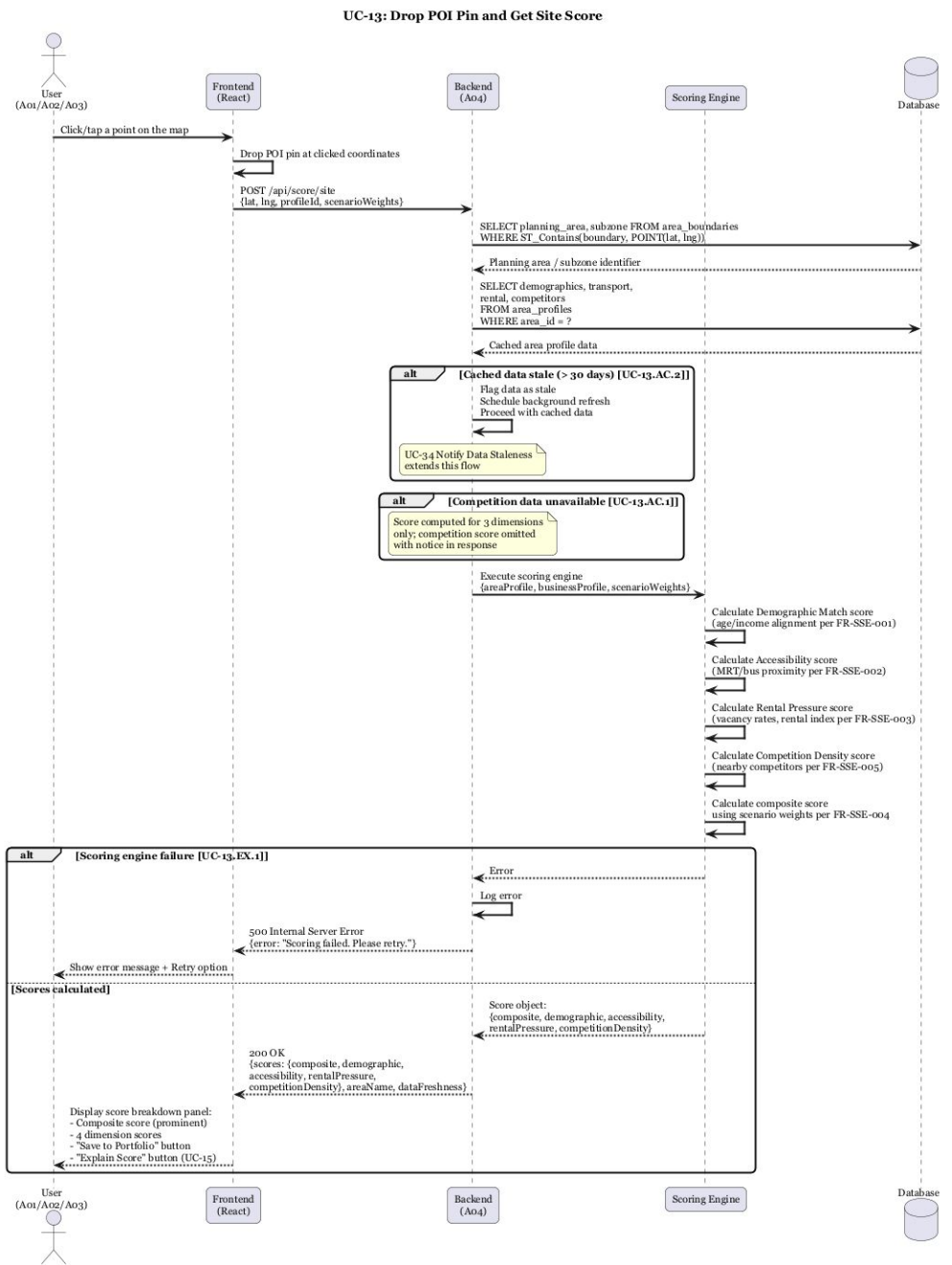


UC-13 Drop POI Pin & Get Site Score



UC-13 Drop POI Pin & Get Site Score

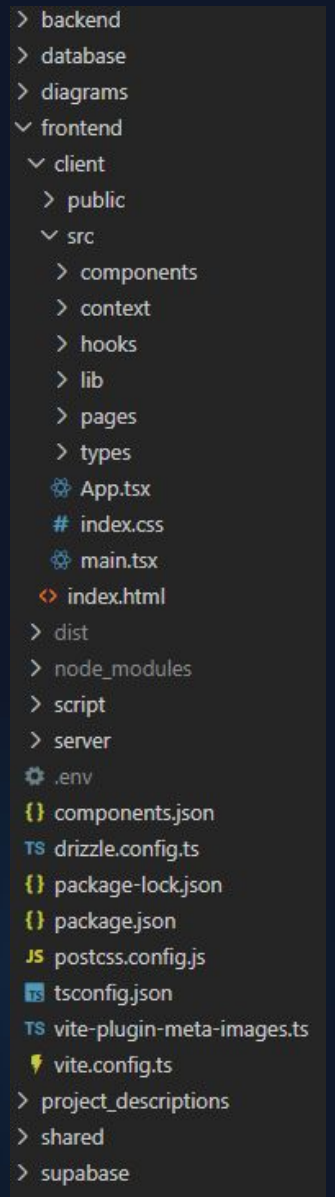




📂 diagrams\Sequence Diagrams

📂 diagrams\Class Diagrams

REPOSITORY STRUCTURE




```
SmartLocateSG/
├── frontend/                    — Frontend app + server
│   ├── package.json             — Frontend dependencies
│   ├── tsconfig.json            — TypeScript config
│   ├── vite.config.ts           — Vite build config
│   ├── client/                  — Single-page app
│   │   ├── public/              — Static public files
│   │   └── src/                 — React source code
│   │       ├── components/      — Reusable UI components
│   │       ├── components/ui/   — UI primitives library
│   │       ├── context/         — React contexts
│   │       ├── hooks/           — Custom React hooks
│   │       ├── lib/             — Client libraries/util
│   │       ├── pages/           — Page-level components
│   │       └── types/           — Type declarations
│   ├── script/                  — Build/helper scripts
│   └── server/                  — Dev API / server shim
│       ├── db/                  — DB helpers/wrappers
│       ├── routes/              — API route handlers
│       └── storage/             — Storage utilities
├── backend/                     — Data ingestion scripts
│   └── scripts/                 — Data fetch & ETL
├── database/                    — DB schemas & data
├── shared/                      — Shared types & schemas
├── supabase/                    — Supabase functions repo
│   └── functions/               — Edge/function handlers
```

GUIDED DEMO SEQUENCE

Phase 1: Setup

Display landing page and walk through Business Profile creation and weighting.

Phase 2: Explore

Drop pins on the map and reveal real-time score breakdowns per location.

Phase 3: Decide

Save top candidates and display side-by-side comparison for final evaluation.

Questions?

SmartLocateSG: Singapore Site Selection intelligence

Group 13 (SCEZ - Group 2)

CHOO J**UNG KIT**

ETHAN VINORAJ

MARCUS LEE JIA RONG

NEO **YU HUI** (LIANG YUHUI)

WAI YAN MIN KO KO